

Deep Learning-Based Building: Footprint Segmentation from Satellite Imagery

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Abstract—In this capstone project, I developed and evaluated a deep learning-based approach for automated building footprint extraction from high-resolution aerial imagery. I implemented and compared three state-of-the-art semantic segmentation models—U-Net, DeepLabV3+, and SegFormer—to identify building regions accurately. The project involved data preprocessing, model training, and performance evaluation using metrics such as Union(IoU), Dice Score, and building count error. I also performed a comparative analysis of the models to assess their segmentation quality and building count estimation capabilities. The results demonstrate the effectiveness of deep learning techniques for large-scale building footprint mapping and provide insights into the strengths of both convolutional and transformer-based architectures for remote sensing applications.

Index Terms—Remote Sensing, Semantic Segmentation, U-Net, DeepLabV3+, SegFormer, Building Extraction, Satellite, IoU Score, Dice Score, Deep Learning

I. INTRODUCTION

A. Background

The rapid growth of urban environments has increased the demand for accurate and up-to-date geographic information. Building footprint maps play an essential role in urban planning, disaster management, infrastructure monitoring, population estimation, and geographic information system (GIS) development. Traditionally, building footprints are generated through manual interpretation of aerial or satellite imagery, a process that is time-consuming, labor-intensive, and difficult to scale across large geographic regions.

Recent advances in deep learning have significantly improved the performance of computer vision tasks, particularly semantic segmentation. Semantic segmentation enables pixel-level classification of images, making it well suited for building footprint extraction from remotely sensed imagery. Modern architectures can automatically learn complex spatial patterns and distinguish buildings from surrounding background features such as roads, vegetation, and open land.

B. Problem Statement

Automatic extraction of building footprints from aerial imagery remains a challenging task due to variations in building size, shape, orientation, illumination conditions, shadows, and surrounding environmental features. Traditional image processing techniques often struggle to handle these variations and typically require extensive manual tuning. Therefore, there is

a need for robust automated methods capable of accurately identifying building structures across diverse scenes while maintaining scalability for large datasets.

C. Objectives

The primary objective of this project is to investigate the effectiveness of deep learning-based semantic segmentation techniques for automated building footprint extraction from high-resolution aerial imagery.

The specific objectives are:

- To develop an end-to-end pipeline for building footprint extraction.
- To implement and train U-Net, DeepLabV3+, and SegFormer architectures.
- To compare the performance of the three models using IoU and Dice Score.
- To evaluate object-level performance through building count estimation.
- To assess model generalization on previously unseen test images.

D. Project Contributions

The main contributions of this project are:

- Development of a complete building footprint extraction framework using deep learning.
- Comparative evaluation of CNN-based and transformer-based segmentation architectures.
- Implementation of sliding-window inference for segmentation of large aerial images.
- Application of connected component analysis for building count estimation.
- Analysis of both pixel-level segmentation accuracy and object-level building counting performance.

II. PREVIOUS WORK

A. U-Net

U-Net was introduced by Ronneberger et al. [2] as a convolutional neural network architecture for biomedical image segmentation. The architecture consists of an encoder-decoder structure with skip connections that transfer spatial information from the encoder to the decoder. These skip

connections enable precise localization while maintaining contextual information, making U-Net highly effective for pixel-level segmentation tasks. Due to its simplicity and strong performance, U-Net has become a widely adopted baseline for remote sensing and building extraction applications.

B. DeepLabV3+

Chen et al. [3] proposed DeepLabV3+, an encoder-decoder architecture that incorporates Atrous Spatial Pyramid Pooling (ASPP). Atrous convolutions allow the network to capture contextual information at multiple scales without significantly increasing computational cost. The ASPP module processes image features using multiple dilation rates, enabling improved segmentation of objects with varying sizes. DeepLabV3+ has demonstrated strong performance across numerous semantic segmentation benchmarks and is particularly effective in complex scenes.

C. Vision Transformers

The introduction of Vision Transformers (ViTs) by Dosovitskiy et al. [4] demonstrated that transformer architectures could achieve competitive performance in image recognition tasks. Unlike convolutional neural networks, transformers utilize self-attention mechanisms to capture long-range dependencies across the entire image. This capability enables improved modeling of global context and has motivated the development of transformer-based segmentation architectures.

D. SegFormer

SegFormer, proposed by Xie et al. [1], combines a hierarchical transformer encoder with a lightweight multilayer perceptron decoder. The architecture eliminates the need for positional encoding and efficiently fuses multi-scale features. SegFormer achieves strong segmentation performance while maintaining computational efficiency, making it attractive for practical applications involving large-scale imagery.

E. Transformer-Based Semantic Segmentation

Transformer-based segmentation architectures have gained significant attention due to their ability to model long-range spatial relationships. Zheng et al. [5] introduced a sequence-to-sequence perspective for semantic segmentation, demonstrating the effectiveness of transformer models in dense prediction tasks. These advances have motivated further exploration of transformer-based methods for remote sensing applications, including building footprint extraction.

The architectures discussed above represent important milestones in semantic segmentation research. In this project, U-Net, DeepLabV3+, and SegFormer were selected for comparison because they represent distinct approaches to feature extraction and contextual understanding, providing an opportunity to evaluate both convolutional and transformer-based methodologies for building footprint extraction.

III. METHODOLOGY

A. Overview

This project aims to automatically extract building footprints from high-resolution aerial imagery using deep learning-based semantic segmentation. The overall workflow consists of dataset preparation, mask generation, patch creation, data augmentation, model training, evaluation, sliding-window inference, and building count analysis. Three segmentation architectures — U-Net[2], DeepLabV3+[3], and SegFormer[1] — were implemented and compared using identical training and evaluation procedures.

B. Dataset

The SpaceNet 2 Paris Buildings dataset [6] was used for this project. The dataset contains high-resolution RGB aerial images along with GeoJSON files containing polygon annotations of building footprints.

Each image is paired with a corresponding GeoJSON file that stores the geographic coordinates of building boundaries. Since deep learning models require pixel-level labels, the polygon annotations were converted into binary segmentation masks.

Dataset Components

- RGB aerial images (.tif)
- Building footprint annotations (.geojson)
- Binary segmentation masks (generated during preprocessing)

The dataset was divided into training and validation subsets using an 80:20 split to ensure unbiased performance evaluation.

C. Mask Generation

The original building annotations are provided as polygons in GeoJSON format. These polygons were rasterized to generate binary masks matching the dimensions of their corresponding aerial images.

For each image:

- 1) The image dimensions and spatial transformation matrix were obtained using Rasterio.
- 2) Building polygons were extracted from the GeoJSON file.
- 3) Rasterization converted the polygons into pixel-level labels.
- 4) Pixels belonging to buildings were assigned a value of 1, while background pixels were assigned a value of 0.

This process transformed geographic vector data into a format suitable for semantic segmentation.

D. Patch Creation and Random Sampling

The original aerial images are large and cannot be processed efficiently as a whole during training. Therefore, a patch-based training strategy [7] [8] was adopted.

Patch Size

- 512×512 pixels

Random Patch Extraction

For every training sample:

- 1) A random location was selected within the image.
- 2) A 512×512 image patch was extracted.
- 3) The corresponding mask patch was extracted from the same location.
- 4) Both image and mask patches were used as a training sample.

Random patch extraction provides:

- Increased data diversity
- Reduced memory consumption
- Better generalization capability

E. Data Augmentation

To improve model robustness and reduce overfitting, online data augmentation [9] [2] was applied during training.

The following augmentations were used:

- Horizontal Flip
- Vertical Flip
- Random Rotation (90°)

These transformations generate additional variations of the training samples while preserving building structures.

F. Data Loading

The processed patches were loaded using PyTorch's Dataset and DataLoader classes.

Training DataLoader

- Batch Size: 4
- Shuffle: Enabled

Validation DataLoader

- Batch Size: 4
- Shuffle: Disabled

The DataLoader efficiently feeds image-mask pairs into the neural network during training and validation.

G. Semantic Segmentation Models

1) *U-Net*: U-Net [2] is a convolutional encoder-decoder architecture designed for pixel-wise image segmentation.

The architecture consists of:

- Contracting encoder path
- Bottleneck layer
- Expanding decoder path
- Skip connections between encoder and decoder levels

Skip connections help preserve spatial information and improve localization accuracy.

2) *DeepLabV3+*: DeepLabV3+ [3] extends conventional encoder-decoder segmentation networks by incorporating Atrous Spatial Pyramid Pooling (ASPP).

Key components include:

- Atrous (dilated) convolutions
- Multi-scale feature extraction
- Encoder-decoder refinement

The ASPP module enables the network to capture contextual information at multiple spatial scales.

3) *SegFormer*: SegFormer [1] is a transformer-based semantic segmentation architecture.

Its main components are:

- Hierarchical Transformer Encoder
- Multi-Layer Perceptron (MLP) Decoder
- Multi-scale feature fusion

Unlike conventional CNN-based models, SegFormer can capture both local and global image context efficiently.

H. Loss Function

A hybrid loss function combining Binary Cross-Entropy (BCE) Loss and Dice Loss [10] was used.

Binary Cross-Entropy Loss

BCE measures pixel-wise classification error between predicted and ground-truth masks.

Dice Loss

Dice Loss focuses on overlap quality between predicted and actual building regions.

Combined Loss

The final loss is computed as:

$$CombinedLoss = BCELoss + DiceLoss \quad (1)$$

This combination improves segmentation performance, especially in class-imbalanced datasets where building pixels occupy a smaller portion of the image.

I. Optimization

The Adam optimizer [11] was used to update network parameters during training.

Hyperparameters

- Optimizer: Adam
- Learning Rate: 0.0001
- Batch Size: 4
- Epochs: 40

Adam was selected because of its fast convergence and effectiveness in deep learning applications.

J. Training Procedure

For each epoch:

- 1) A batch of image patches was loaded.
- 2) Forward propagation generated predicted masks.
- 3) Combined loss was computed.
- 4) Backpropagation calculated gradients.
- 5) Adam optimizer updated model weights.
- 6) Validation performance was evaluated after each epoch.

The model achieving the best validation performance was saved for subsequent testing and inference.

K. Evaluation Metrics

1) *Intersection over Union (IoU)*: IoU measures the overlap between predicted and ground-truth masks.

Formula

$$IoU = \frac{Intersection}{Union} \quad (2)$$

Higher IoU values indicate better segmentation performance.

2) *Dice Coefficient*: The Dice Coefficient evaluates similarity between predicted and actual masks.

Formula

$$Dice = \frac{2 \times Intersection}{Prediction + GroundTruth} \quad (3)$$

Dice is particularly useful for binary segmentation tasks.

L. Sliding Window Inference

Since full-sized aerial images exceed GPU memory limitations, sliding-window inference [12] was used during testing.

Procedure

- 1) The image was divided into overlapping **512 × 512 patches**.
- 2) Each patch was processed independently by the trained model.
- 3) Predicted patches were placed back into their original positions.
- 4) Overlapping regions were averaged.
- 5) A complete segmentation mask was reconstructed.

This approach enables segmentation of large images while maintaining prediction quality.

M. Building Count Analysis

Beyond pixel-level segmentation, building counting was performed to evaluate practical usability.

Connected Component Analysis [13] was applied to the predicted masks:

- 1) Connected building regions were identified.
- 2) Small noisy regions were removed.
- 3) Remaining connected components were counted as individual buildings.

The predicted building counts were compared with ground-truth counts to assess object-level performance.

N. Evaluation Strategy

The final stage involved evaluating U-Net, DeepLabV3+, and SegFormer using:

- IoU
- Dice Score
- Building Count Accuracy
- Visual Quality of Segmentation Masks

IV. RESULTS

A. Overview

The performance of the proposed building footprint extraction framework was evaluated using three semantic segmentation architectures: U-Net, DeepLabV3+, and SegFormer. The models were assessed using Intersection over Union (IoU) and Dice Score, which measure the overlap between predicted and ground-truth building masks. In addition, qualitative visual analysis and building count estimation were performed to evaluate the practical effectiveness of each model.

B. Segmentation Performance

The primary objective of the models was to accurately identify building footprints from high-resolution aerial imagery. Segmentation performance was evaluated using IoU and Dice Score on the validation dataset.

TABLE I
SEGMENTATION PERFORMANCE COMPARISON

Model	IoU	Dice Score
U-Net	0.6256	0.7438
DeepLabV3+	0.7254	0.8289
SegFormer	0.7063	0.8165

Table I presents the IoU and Dice Score obtained by the three segmentation models on the validation dataset. DeepLabV3+ achieved the highest IoU (0.7254) and Dice Score (0.8289), followed by SegFormer and U-Net.

C. Qualitative Segmentation Results

While numerical metrics provide a quantitative assessment of model performance, visual inspection offers additional insight into segmentation quality. Representative validation samples were selected to compare the predictions produced by each architecture.

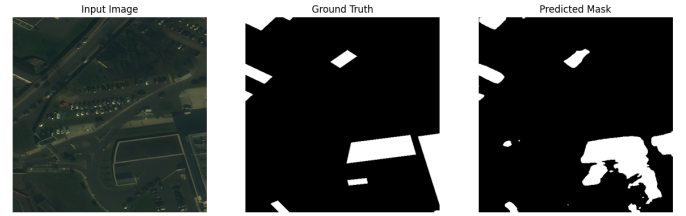


Fig. 1. U-Net: Ground Truth vs Predicted Mask

1) *U-Net Segmentation*: The U-Net model successfully identified most building regions and generated relatively accurate building boundaries. However, some inaccuracies were observed around small structures and densely packed urban areas.

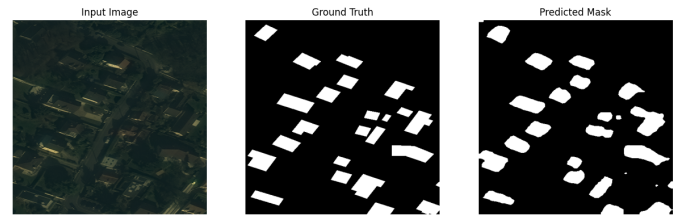


Fig. 2. DeepLabV3+: Ground Truth vs Predicted Mask

2) *DeepLabV3+ Segmentation*: DeepLabV3+ produced the most accurate segmentation masks among the evaluated models. The predictions exhibited strong boundary delineation and improved detection of buildings with varying sizes and shapes.

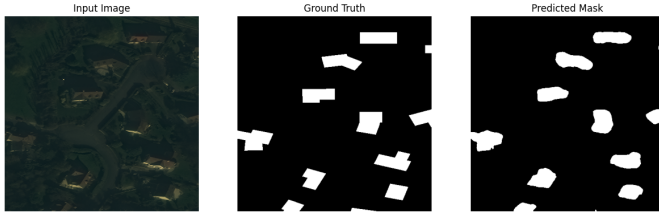


Fig. 3. SegFormer: Ground Truth vs Predicted Mask

3) *SegFormer Segmentation*: SegFormer generated high-quality segmentation masks and effectively preserved building structures in complex scenes. The model demonstrated strong contextual understanding and produced fewer fragmented predictions compared to conventional CNN-based approaches.

D. Sliding Window Inference

The original aerial images are significantly larger than the training patch size of 512×512 pixels. To perform inference on full-resolution imagery, a sliding-window approach with overlapping patches was employed. Predictions from overlapping regions were averaged and merged to reconstruct complete building footprint masks.

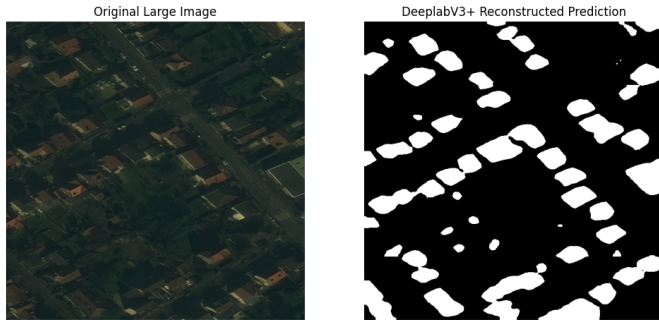


Fig. 4. DeepLabV3+ Reconstruction

The reconstructed segmentation masks showed smooth transitions across patch boundaries and preserved the spatial continuity of building structures. This demonstrates the effectiveness of the sliding-window approach for large-scale aerial image segmentation without exceeding GPU memory limitations.

E. Building Count Estimation

To evaluate the practical usefulness of the generated segmentation masks, Connected Component Analysis was applied to estimate the number of buildings present in each image. The predicted building counts were compared against ground-truth counts, and the mean counting error was calculated.

Table II indicates that SegFormer achieved the lowest mean building count error of 2.0174 buildings per image, followed by U-Net (2.2087) and DeepLabV3+ (3.2043).

TABLE II
BUILDING COUNT ESTIMATION

Model	Mean Count Error
U-Net	2.2087
DeepLabV3+	3.2043
SegFormer	2.0174

F. Model Comparison on a Common Test Image

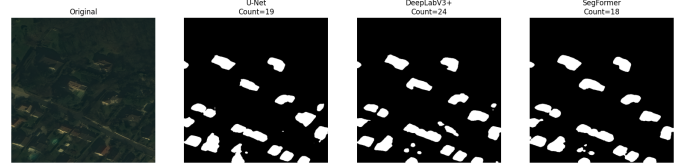


Fig. 5. Comparison of model predictions on a common test image

This experiment was conducted to visually compare the behavior of all three architectures on the same aerial image. The original image is shown alongside the segmentation masks produced by U-Net, DeepLabV3+, and SegFormer. Building counts estimated through Connected Component Analysis are displayed above each prediction.

For this sample image, U-Net predicted 19 buildings, DeepLabV3+ predicted 24 buildings, and SegFormer predicted 18 buildings. All three models successfully identified the major building structures present in the scene. Visual differences can be observed in the segmentation masks and the resulting building counts.

This visual comparison supports the quantitative findings presented earlier. DeepLabV3+ demonstrated strong segmentation capability, while SegFormer produced more coherent object-level predictions that contributed to its lower building count error.

G. Generalization on Unseen Test Images

To evaluate the ability of the trained models to generalize beyond the validation dataset, predictions were generated on previously unseen test images. These images were not used during training or validation and therefore provide an indication of real-world deployment performance.

Figures 6–10 present predictions generated on previously unseen test images. The examples include both building-rich scenes and images containing no buildings.

H. Summary of Results

The experimental evaluation demonstrated that all three architectures successfully extracted building footprints from aerial imagery. DeepLabV3+ achieved the highest IoU and Dice Score, while SegFormer achieved the lowest building count error. Predictions generated on unseen test images further demonstrated the practical applicability of the trained models.

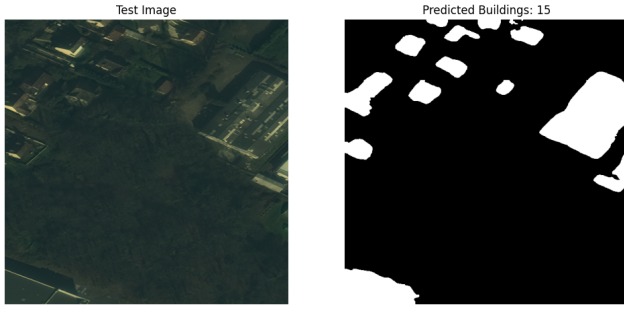


Fig. 6. Test Image 1. The model successfully detected 15 buildings and produced coherent building footprints despite variations in illumination and background appearance.

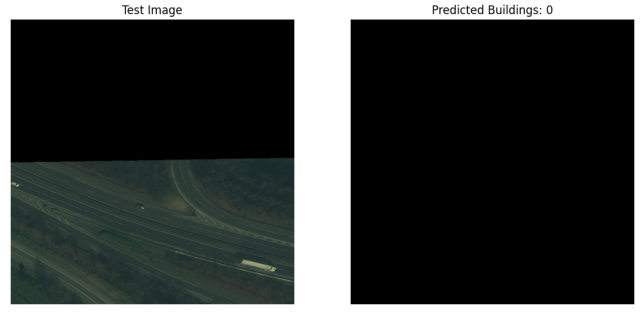


Fig. 9. Test Image 5. Prediction generated on an unseen test image containing primarily transportation infrastructure. The model correctly predicted zero buildings, indicating strong resistance to false building detections.

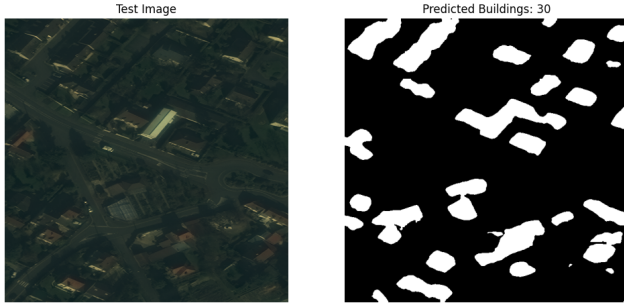


Fig. 7. Test Image 4. The model detected 30 buildings and successfully segmented densely distributed structures, demonstrating robustness in complex urban environments.

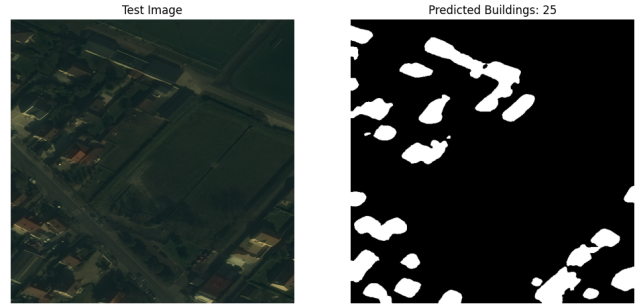


Fig. 10. Test Image 3. The model identified 25 buildings and accurately segmented the majority of visible structures, maintaining well-defined boundaries despite variations in building size and orientation.

V. DISCUSSION

A. Segmentation Performance Analysis

The experimental results indicate that DeepLabV3+ achieved the highest segmentation accuracy among the evaluated architectures, obtaining an IoU of 0.7254 and a Dice Score of 0.8289. The superior performance of DeepLabV3+ can largely be attributed to its Atrous Spatial Pyramid Pooling (ASPP) module, which captures contextual information at multiple spatial scales. Since buildings within aerial imagery vary considerably in size and shape, the ability to extract multi-scale features provides a significant advantage.

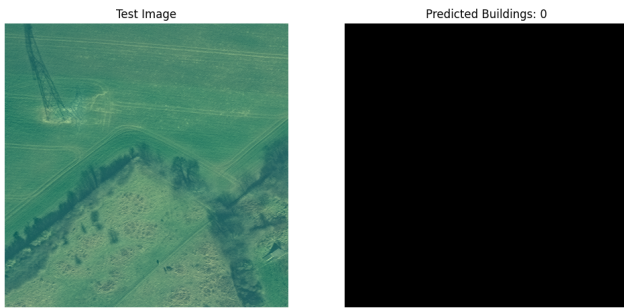


Fig. 8. Test Image 2. Prediction generated on an unseen test image containing no visible building structures. The model correctly predicted zero buildings, demonstrating its ability to distinguish between built and non-built environments while minimizing false positives.

SegFormer achieved competitive segmentation performance with an IoU of 0.7063 and a Dice Score of 0.8165. Although its segmentation metrics were slightly lower than those of DeepLabV3+, the transformer-based architecture demonstrated strong contextual understanding and effective feature representation. U-Net achieved an IoU of 0.6256 and a Dice Score of 0.7438, providing a solid baseline despite its comparatively simpler architecture.

B. Building Count Analysis

While DeepLabV3+ achieved the best pixel-level segmentation accuracy, SegFormer produced the lowest building count error. This suggests that segmentation quality alone does not fully capture practical performance in building extraction tasks. Building counting depends heavily on preserving individual building boundaries and correctly separating adjacent structures.

The transformer-based feature extraction employed by SegFormer appears to provide advantages in maintaining object-level consistency. As a result, SegFormer was more effective at generating segmentation masks that translated into accurate building count estimates through connected component analysis.

C. Comparison with Initial Expectations

Prior to conducting the experiments, I expected SegFormer to outperform the CNN-based architectures due to the recent

success of transformer models in computer vision and semantic segmentation tasks. Transformers have demonstrated strong performance in capturing long-range dependencies and global contextual information, which appeared highly relevant for building extraction from aerial imagery.

However, the experimental results revealed that DeepLabV3+ achieved the highest IoU and Dice scores. This finding suggests that multi-scale convolutional feature extraction remains highly effective for building footprint segmentation, particularly on the SpaceNet Paris dataset. At the same time, SegFormer achieved the best building counting performance, partially validating the expectation that transformer-based architectures better preserve object-level information.

D. Limitations

Several limitations should be considered when interpreting the results of this study.

- The dataset is limited to a single geographic region (Paris), which may restrict model generalization to other cities and environments.
- The study focuses on binary semantic segmentation and does not explicitly separate touching or overlapping buildings.
- Building counting relies on connected component analysis, which may merge adjacent structures or split large buildings into multiple components.
- Computational resource limitations restricted experimentation with larger model variants and extensive hyperparameter optimization.

Future studies could address these limitations by incorporating multi-city datasets, instance segmentation techniques, and larger transformer architectures.

E. Key Findings

The main findings of this study can be summarized as follows:

- DeepLabV3+ achieved the highest segmentation accuracy with an IoU of 0.7254 and a Dice Score of 0.8289.
- SegFormer achieved the lowest building count error, demonstrating strong object-level performance.
- U-Net provided a reliable baseline and remained competitive despite its simpler architecture.
- The trained models generalized effectively to unseen test imagery.
- Both convolutional and transformer-based architectures demonstrated strong potential for automated building footprint extraction from aerial imagery.

The results indicate that no single architecture was universally superior across all evaluation criteria. While DeepLabV3+ achieved the best pixel-level segmentation performance, SegFormer demonstrated stronger object-level performance through building counting. This highlights the importance of evaluating semantic segmentation models using multiple metrics rather than relying solely on IoU or Dice Score.

VI. CONCLUSION & FUTURE WORK

In this project, I investigated the use of deep learning-based semantic segmentation techniques for automated building footprint extraction from high-resolution aerial imagery. I implemented, trained, and evaluated three state-of-the-art architectures—U-Net, DeepLabV3+, and SegFormer—using the SpaceNet 2 Paris Buildings dataset. The project encompassed the complete workflow of data preprocessing, mask generation, patch-based training, semantic segmentation, sliding-window inference, and building count estimation.

The experimental results demonstrated that all three models were capable of accurately identifying building footprints from aerial imagery. Quantitative evaluation using Intersection over Union (IoU) and Dice Score showed that DeepLabV3+ achieved the best segmentation performance, obtaining an IoU of 0.7254 and a Dice Score of 0.8289. These results highlight the effectiveness of Atrous Spatial Pyramid Pooling in capturing multi-scale contextual information and improving pixel-level segmentation accuracy.

Building count estimation provided an additional object-level evaluation of model performance. SegFormer achieved the lowest mean building count error, demonstrating its ability to preserve individual building structures and accurately separate adjacent buildings. Although its segmentation metrics were slightly lower than those of DeepLabV3+, SegFormer exhibited strong practical performance for downstream building analysis tasks.

Through qualitative analysis, I observed that the generated segmentation masks closely matched the ground-truth building footprints. The sliding-window inference approach successfully enabled prediction on large aerial images without exceeding memory constraints. Furthermore, evaluation on previously unseen test images showed that the models generalized well beyond the validation dataset, accurately detecting buildings in urban scenes while avoiding false detections in images containing no buildings.

Overall, I demonstrated that deep learning-based semantic segmentation provides an effective and scalable solution for automated building footprint extraction. DeepLabV3+ emerged as the strongest model for pixel-level segmentation accuracy, while SegFormer achieved the best object-level building counting performance. These findings highlight the potential of modern semantic segmentation architectures for remote sensing applications such as urban planning, infrastructure monitoring, disaster management, and geographic information system (GIS) development.

In future work, I plan to train and evaluate the models on datasets from multiple geographic regions to improve generalization. Additional improvements may include exploring larger transformer-based architectures, incorporating instance segmentation techniques to separate touching buildings, and evaluating the framework on additional remote sensing datasets. These enhancements could further improve the robustness and practical applicability of automated building footprint extraction systems.

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